

Generation Interconnection System Impact Study Report

For Queue Project Q20-1499

Western Michigan University Power Plant 100 kV
June 27, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1499. The Interconnection Customer proposes a 174.8 MW solar photovoltaic facility interconnecting at the Western Michigan University Power Plant 100 kV substation in the Michigan area. The request was received on June 15, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 551 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,184,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Parkview Battery / Western Michigan University Power Plant 100 KV, the Western Michigan University Power Plant / Galt Substation 100 KV loads to 113.8% of its 132 MVA rating. Project DFAX: 100.0%.

The Parkview Battery / Western Michigan University Power Plant 100 KV is overloaded to 107.2% of its 140 MVA emergency rating for the loss of the Western Michigan University Power Plant / Galt 100 KV. The project contributes 100.0% DFAX (174.8 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Western Michigan University Power Plant / Galt Substation 100 KV	Rebuild line with higher-capacity conductor	\$2,631,000	100% (\$2,631,000)
Parkview Battery / Western Michigan University Power Plant 100 KV	Reconductor line	\$553,000	100% (\$553,000)
	Total allocated to project		\$3,184,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.